HOMEWORK 2.8 · UNIT 2 · NC.8.EE.3 · NC.8.EE.4

Scientific Notation Practice

Targets: EE.3-2, EE.4-1, EE.4-2, EE.3-4, EE.3-5, EE.4-5, EE.4-3, EE.4-7, EE.4-8

Name _____ Date _____ Block _____

Remember: Multiply: multiply the coefficients, **add** the exponents. **Divide:** divide the coefficients, **subtract** the exponents. Then fix the form: $1 \le$ coefficient < 10. Example: $(6 \times 10^4)(5 \times 10^2) = 30 \times 10^6 = 3 \times 10^7$.

SCORE 3.0 Approaching · the basics

Targets: EE.3-2, EE.4-1, EE.4-2, EE.3-4, EE.3-5

1 Write in standard form. REVIEW 2.6 EE.3-2

8.1×10^{-5}

2 Multiply. $(4 \times 10^5)(7 \times 10^{-2})$ EE.4-1

3 Divide. $(3.6 \times 10^7) \div (1.2 \times 10^{-2})$ EE.4-2

4 Circle >, <, or =. 5×10^8 ___ 2×10^9 EE.3-4,EE.3-5

(>) (<) (=)

Why? _____

5 A blue whale has a mass of about 1.5×10^5 kg. A dog has about 3×10^1 kg. How many times as heavy is the whale? EE.3-4,EE.4-2

(A) 5,000 times (B) 500 times (C) 50,000 times (D) 2 times

SCORE 4.0 Proficient · on target

Targets: EE.3-5, EE.4-5, EE.4-1, EE.4-3

6 Distance from the Sun: EE.3-5

Planet	km
Jupiter	7.8×10^8
Mercury	5.8×10^7
Neptune	4.5×10^9
Earth	1.5×10^8
Mars	2.3×10^8

Which list goes from **least to greatest**?

(A) Mercury, Earth, Mars, Jupiter, Neptune
(B) Earth, Mars, Neptune, Mercury, Jupiter
(C) Neptune, Jupiter, Mars, Earth, Mercury
(D) Mercury, Jupiter, Earth, Mars, Neptune

7 A school uses 1.5×10^4 sheets of paper a week. How many sheets in 36 weeks? Use scientific notation. EE.4-5

8 **Sam** made a mistake. Circle it. Explain it. Fix it. REVIEW 2.7 EE.4-1,EE.4-3

SAM'S WORK
$(5 \times 10^6)(6 \times 10^3) = 30 \times 10^{18}$

SCORE 5.0 Mastery · go deeper

Targets: EE.4-7, EE.4-8

9 A heart beats about 1×10^5 times a day. A year has about 3.65×10^2 days. About how many beats is that in a year? Justify each step. EE.4-7

I multiply _____ and add _____ . The answer is _____ .

10 Luis says a grain of salt has a mass of 6.5×10^5 grams. Is that reasonable? What mistake did he likely make? EE.4-8

It is _____ because _____ . He likely _____ .